



Semester 2 Examinations 2018/2019

Exam Code(s)	3BA1, 3BME1, 3BMS2, 3BS9
Exam(s)	Bachelor of Science (Hons.) Bachelor of Science (Mathematical Science) Honours Bachelor of Arts
Modules	Fields and Applications
Module Codes	MA3491
Paper	1
External Examiner(s)	Prof. Thomas Brady
Internal Examiner(s)	Prof. Graham Ellis Dr Emil Sköldbberg
<u>Instructions:</u>	Students should attempt ALL questions. (All questions carry equal marks).
Duration	2 Hours
No. of Pages	3 (including this page)
Discipline	Mathematics

1. (a) Let A be an alphabet. Define the *Hamming distance* on the set A^n , and show that the Hamming distance is a metric. [7 marks]
- (b) State the *sphere packing bound* and use it to show that a binary code of block length 8 that can correct all single errors can have at most 28 code words [7 marks]
- (c) In the following ISBN numbers, one symbol has been replaced by a '#'. Find the missing symbol.
 - (i) 0-201-10#42-7 [3 marks]
 - (ii) 978-0-470-65#84-7 [3 marks]
- (d) Consider the code C over the alphabet $\{A, B\}$. Determine the minimum distance of C if it consists of the codewords

$$\{ABABA, BABAB, AAABB, BBBAB\}.$$

[5 marks]

2. (a) Find the greatest common divisor of the polynomials $x^3 + 2x^2 + 2x + 1$ and $x^3 + x + 1$ in $\mathbb{Z}_3[x]$. [5 marks]
- (b) Show that the ternary polynomial $x^3 + 2x + 1$ is irreducible. [5 marks]
- (c) Let R be a Euclidean domain.
 - (i) State what it means that an element of R is a *unit*. [2 marks]
 - (ii) Show that $\mathcal{U}(R)$, the set of units of R , forms a group under multiplication. [4 marks]
 - (iii) Let F be a field. What are the elements of $\mathcal{U}(F[x])$? Motivate your answer fully. [2 marks]
- (d) Construct the field of eight elements. (I.e. give its elements and show how they are added and multiplied. You do not have to give the full addition and multiplication tables). [7 marks]

P.T.O

3. (a) Let C be the binary linear code with generator matrix

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

(i) List the codewords of C . **[4 marks]**

(ii) Hence find the minimum distance of C . **[3 marks]**

- (b) Let C be the ternary code with generator matrix

$$\begin{pmatrix} 0 & 2 & 2 & 1 \\ 1 & 1 & 2 & 0 \end{pmatrix}$$

(i) Find a generator matrix in standard form for C . **[4 marks]**

(ii) Hence, construct a parity check matrix for C . **[3 marks]**

(iii) Construct a syndrome lookup table for C and use it to decode the words 0121 and 2212. **[6 marks]**

4. (a) (i) What is the definition of the Hamming code $Ham(r, q)$? **[2 marks]**

(ii) Write down a parity check matrix for the Hamming code $Ham(2, 5)$. **[6 marks]**

(iii) Explain how a single error can be corrected in $Ham(2, 5)$. Decode the words 101023 and 040404. **[5 marks]**

- (b) Let C be the ternary cyclic code of length 4 with generator polynomial $x^2 - 1$.

(i) Give the generator matrix of C . **[4 marks]**

(ii) Find the check polynomial of C . **[4 marks]**

(iii) Write down the check matrix of C and hence verify that 2211 is a codeword in C . **[4 marks]**